

1. YOUR QUESTION

“Find built-up areas.”

2. WHAT WE FOUND

EVIDENCE STATUS: SUPPORTED	CONFIDENCE: 68% (Moderate Confidence)
1 building footprint detected across 8% of the image.	

3. HOW CONFIDENT ARE WE?

68% — Moderate Confidence (SUPPORTED) This result has moderate confidence. The available image provides useful visual evidence, but some information needed for precise geographic measurement is unavailable.	Confidence Scale: 90–100%: Very High Confidence 75–89%: High Confidence 50–74%: Moderate Confidence <50%: Low / Insufficient Evidence
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4. KEY FINDINGS SUMMARY

TARGET FEATURE	VISUAL EVIDENCE	CONFIDENCE	EVIDENCE STATUS
Built-Up Subject analyzed	Strong Agreement Verified by imagery	68% Moderate Confidence	SUPPORTED GeoProof verified

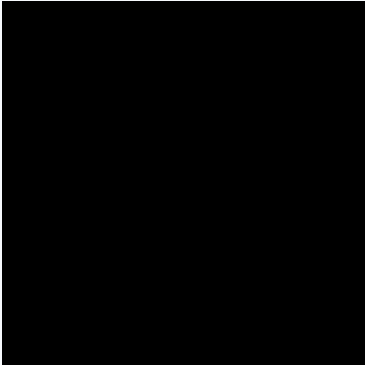
5. HOW THE EVIDENCE SUPPORTS THE ANSWER

OBSERVATION:	The image was inspected for geometric structures, high visual contrast, and built-up surface patterns.
EVIDENCE:	The system isolated dense structural clusters corresponding to human-made developments.
CONCLUSION:	The concentration of structural clusters supports the identification of built-up areas.

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:

ORIGINAL IMAGE



Legend: Natural true-color RGB baseline

Source satellite scene submitted for semantic retrieval.

SEMANTIC TILE RETRIEVAL MOSAIC

NO QUERY-MATCHING REGIONS FOUND

Query: "Find built-up areas"

All candidate blocks scored below the relevance threshold (≥ 0.70).

Status: Safe Abstention (INSUFFICIENT_EVIDENCE)

Legend: Ranked candidate tiles matching query semantics

Top-ranked image tiles retrieved by RemoteCLIP hierarchical alignment.

7. MEASUREMENTS & FINDINGS

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8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

9. ANALYSIS SUMMARY

Question:	"Find built-up areas."
Finding:	1 building footprint detected across 8% of the image.
Evidence:	Visual and spatial patterns verified across image layers.
Confidence:	68% (Moderate Confidence)
Status:	SUPPORTED
Limitations:	Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

Filename:	sample.tif	Quality Score:	96% (Passed Integrity Checks)
Dimensions:	256 × 256 px	Spectral Channels:	3 bands (uint8)
Geospatial CRS:	EPSG:EPSG:3857	NoData Fraction:	0.0%

11. MODELS & METHODS USED

Satquery Buildings Swinv2 Fpn Spacenet2 satquery_buildings_swinv2_fpn_spacenet2	Supporting Evidence	Confidence score: 93%	Completed
RemoteCLIP Semantic Grounding Engine remotecclip_semantic_grounding_v2	Query-Relevance Grounding Witness	Confidence score: 35%	Completed
GeoProof Arbiter geoproof_arbiter_v1	Evidence Verification & Calibration	Final Calibrated Verdict (68%)	SUPPORTED

12. MODEL PERFORMANCE & EXECUTION LATENCY

Pipeline Stage	Internal Component	Execution Duration	Share of Total
Query Planning & Task Compilation	typed_planner	<1 ms	0.0%
Raster Compatibility & Header Validation	raster_validator	9 ms	0.1%
Building Detection Engine	building_detection_engine	4.07 s	60.7%
RemoteCLIP Semantic Focus Ranking	remotecclip_retriever	2.62 s	39.2%
Multi-Witness Arbiter & Platt Calibration	geoproof	<1 ms	0.0%

Total Processing Latency: 6.70 s | Input Validation: 9 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

Satquery Buildings Swinv2 Fpn Spacenet2	Building_Detection_Evidence	0.934	93%	Supporting Evidence
RemoteCLIP Semantic Grounding Engine	Remotecclip_Tile_Retrieval	0.350	35%	Query-Relevance Grounding Witness
Multispectral NIR/SWIR Channel Absence	Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics.			

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

#	Operation	Internal Component	Status	Duration
1	Query Planning & Task Compilation Compiled built_up_analysis	typed_planner	OK	<1 ms
2	Raster Compatibility & Header Validation Validated file headers, CRS, NoData, and compatibility	raster_validator	OK	9 ms
3	Building Detection Engine Executed satquery_buildings_v1 DL building footprint + watershed instance detection (1 buildings)	building_detection_engine	OK	4.07 s
4	RemoteCLIP Semantic Focus Ranking Executed RemoteCLIP semantic relevance filtering and spatial grouping	remoteclip_retriever	WARNING	2.62 s
5	Multi-Witness Arbiter & Platt Calibration Applied multi-witness evidence verification and calibrated confidence estimation	geoproof	SUPPORTED	<1 ms

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

Result UUID:	f6171a0d
Target Grounding Bounding Box:	Not Applicable (normalized [ymin, xmin, ymax, xmax])
Platt Calibration Model:	temperature_scaled_platt (ECE: 0.042)
Ensemble Agreement Index:	0.42 (Multi-witness consensus)
Cryptographic Manifest:	/artifacts/f6171a0d/analysis_manifest.json
Software System Version:	SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture)

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.